

Fall 1991

Problem 1 (Fa79, Sp88, Fa91). Prove that every finite group of order > 2 has a nontrivial automorphism.

Problem 2 (Fa91). Let f be a continuous function from $\mathbb{R}$ to $\mathbb{R}$ such that $|f(x) - f(y)| \geq |x - y|$ for all x and y . Prove that the image of f is all of $\mathbb{R}$.

Note: See also ??.

Problem 3 (Fa91). Evaluate the integral $I = \frac{1}{2\pi i} \int_C \frac{z^{n-1}}{3z^n - 1} dz$ where n is a positive integer, and C is the circle $|z| = 1$, with counterclockwise orientation.

Problem 4 (Fa91). 1. Prove that any real $n \times n$ matrix M can be written as $M = A + S + cI$, where A is antisymmetric, S is symmetric, c is a scalar, I is the identity matrix, and $\text{tr } S = 0$.

2. Prove that with the above notation,

$$\text{tr}(M^2) = \text{tr}(A^2) + \text{tr}(S^2) + \frac{1}{n}(\text{tr } M)^2$$

Problem 5 (Sp90, Fa91). Let f be an infinitely differentiable function from $\mathbb{R}$ to $\mathbb{R}$. Suppose that, for some positive integer n ,

$$f(1) = f(0) = f'(0) = f''(0) = \cdots = f^{(n)}(0) = 0$$

Prove that $f^{(n+1)}(x) = 0$ for some x in $(0, 1)$.

Problem 6 (Fa91). Let the function f be analytic in the disc $|z| < 1$ of the complex plane. Assume that there is a positive constant M such that

$$\int_0^{2\pi} |f'(re^{i\theta})| d\theta \leq M \quad \text{for } 0 \leq r < 1$$

Prove that

$$\int_{[0,1)} |f(x)| dx < \infty$$

Problem 7 (Fa91). Consider the vector differential equation

$$\frac{dx(t)}{dt} = A(t)x(t)$$

where A is a smooth $n \times n$ function on $\mathbb{R}$. Assume A has the property that $\langle A(t)y, y \rangle \leq c\|y\|^2$ for all y in $\mathbb{R}^n$ and all t , where c is a fixed real number. Prove that any solution $x(t)$ of the equation satisfies $\|x(t)\| \leq e^{ct}\|x(0)\|$ for all $t > 0$.

Problem 8 (Fa91). Let $a_1, a_2, a_3, \dots$ be positive numbers.

1. Prove that $\sum a_n < \infty$ implies $\sum \sqrt{a_n a_{n+1}} < \infty$.
2. Prove that the converse of the above statement is false.

Problem 9 (Fa91). Let G be a group of order $2p$, where p is an odd prime. Assume that G has a normal subgroup of order 2. Prove that G is cyclic.

Problem 10 (Su79, Sp82, Sp83, Su84, Fa91, Fa98, Sp11, Fa15, Fa16, Fa23). Let $\mathbb{F}$ be a finite field with q elements and let V be an n -dimensional vector space over $\mathbb{F}$.

1. Determine the number of elements in V .
2. Let $\text{GL}_n(\mathbb{F})$ denote the group of all $n \times n$ nonsingular matrices over $\mathbb{F}$. Determine the order of $\text{GL}_n(\mathbb{F})$.
3. Let $\text{SL}_n(\mathbb{F})$ denote the subgroup of $\text{GL}_n(\mathbb{F})$ consisting of matrices with determinant 1. Find the order of $\text{SL}_n(\mathbb{F})$.

Problem 11 (Fa91). Let X and Y be metric spaces and f a continuous map of X into Y . Let $K_1, K_2, \dots$ be nonempty compact subsets of X such that $K_{n+1} \subset K_n$ for all n , and let $K = \bigcap K_n$. Prove that

$$f(K) = \bigcap f(K_n)$$

Note: See also ??.

Problem 12 (Fa91). Let p be a nonconstant complex polynomial whose zeros are all in the half-plane $\Im z > 0$.

1. Prove that $\Im(p'/p) > 0$ on the real axis.

2. Find a relation between $\deg p$ and

$$\int_{-\infty}^{\infty} \Im \frac{p'(x)}{p(x)} dx$$

Problem 13 (Fa91). Let $A = (a_{ij})_{i,j=1}^n$ be a real $n \times n$ matrix with nonnegative entries such that

$$\sum_{j=1}^n a_{ij} = 1 \quad 1 \leq i \leq n$$

Prove that no eigenvalue of A has absolute value greater than 1.

Problem 14 (Fa91). Let $\mathcal{B}$ denote the unit ball of $\mathbb{R}^3$, $\mathcal{B} = \{r \in \mathbb{R}^3 \mid \|r\| \leq 1\}$. Let $J = (J_1, J_2, J_3)$ be a smooth vector field on $\mathbb{R}^3$ that vanishes outside of $\mathcal{B}$ and satisfies $\nabla \cdot \vec{J} = 0$.

1. For f a smooth, scalar valued function defined on a neighborhood of $\mathcal{B}$, prove that

$$\int_{\mathcal{B}} (\nabla f) \cdot \vec{J} dx dy dz = 0$$

2. Prove that

$$\int_{\mathcal{B}} J_1 dx dy dz = 0$$

Problem 15 (Fa91). Let $\mathfrak{I}$ be the ideal in the ring $\mathbb{Z}[x]$ generated by $x - 7$ and 15. Prove that the quotient ring $\mathbb{Z}[x]/\mathfrak{I}$ is isomorphic to $\mathbb{Z}_{15}$.

Problem 16 (Fa91). Let $M_{n \times n}$ be the space of real $n \times n$ matrices. Regard it as a metric space with the distance function

$$d(A, B) = \sum_{i,j=1}^n |a_{ij} - b_{ij}| \quad A = (a_{ij}), B = (b_{ij})$$

Prove that the set of nilpotent matrices in $M_{n \times n}$ is a closed set.

Problem 17 (Fa91). Let f be a C^1 function from the interval $(-1, 1)$ into a number ε in $(0, 1)$ such that $\|f(t)\|$ is an increasing function of t on $(0, \varepsilon)$.

Problem 18 (Fa91). Let the function f be analytic in the entire complex plane and satisfy the inequality $|f(z)| \leq |\Re z|^{-1/2}$ off the imaginary axis. Prove that f is constant.